Instructions

To successfully complete this assessment, make sure to follow these instructions:

- Carefully read and understand each question before answering it.
- Respond to ALL questions in the assessment.
- Explicitly display your work and outline the steps taken for each answer. Support your explanations and responses with examples to showcase a comprehensive understanding. Partial marks will be granted for accurate work, even if the final answer is incorrect.
- Take note of the indicated weighting for each question, denoted in parentheses () at the conclusion of each question.

Submission

- Submit your completed assessment online using the following guidelines:
- Name and save your files as "CBS131_Report1_LastName_FirstName"
- Utilize the Assessment Submission link available under the Assessment tab in the primary navigation menu.
- Access your Learning Facilitator's feedback through the Grade Centre in the Learning Management System (LMS) portal under My Grades.

Learning Outcomes

This assignment covers the following learning outcomes in component CBS131:

a. Demonstrate an understanding of fundamental cybersecurity principles in the contemporary ICT environments
b. Describe risk management strategies in consideration of the emerging cybersecurity threat landscape
c. Discuss a cybersecurity solution utilising available appropriate software, hardware, and network security tools and solutions
d. Report on the impacts of cybersecurity solutions in a relevant business context

Assignment Overview

This individual assignment requires you to design, implement, and document a secure site-to-site VPN solution using IPsec to connect an online gaming company's headquarters (HQ) with a remote data centre. Additionally, you will produce a 1000-word cybersecurity report advising on how to assess the cybersecurity threats facing the online gaming industry and apply an incident response plan to mitigate such attacks.

Scenario

GameSecure Inc., an online gaming company, operates both a headquarters and a data center in different locations. To ensure secure communication between these two sites, GameSecure Inc. has decided to implement a Site-to-Site VPN using IPsec. This will allow encrypted data transmission over the internet, protecting sensitive information from cyber threats. As a cybersecurity expert, you are tasked with designing this network and assessing potential cybersecurity threats to the online gaming industry.

Assignment Task:

You are required to design and implement a secure network for GameSecure Inc. that connects their headquarters to their data centre via an IPsec Site-to-Site VPN. This task will be completed using Cisco Packet Tracer, and you will submit both a technical implementation and a cybersecurity report.

Assignment Structure:

Part 1: VPN Implementation

1. Cisco Packet Tracer File Submission – (35%)

- Submit the Cisco Packet Tracer file that corresponds with your network design. This file should demonstrate a fully functional IPsec Site-to-Site VPN, with all required devices configured according to the requirement.

Part 2: Technical Report

Your report should include the following sections:

2. Introduction - (10%)

- Importance of IPsec VPNs for online gaming companies.
- Overview of the main cybersecurity threats facing the industry.

3. Network Design & Implementation (25%)

- **Network Diagram:** Create a network diagram using Cisco Packet Tracer that includes at least 03 routers, 03 Ethernet switches, and 03 PCs. This should show how the headquarters and data center are connected via the VPN.
- **Configuration Steps:** Provide a detailed, step-by-step explanation of how you set up the IPsec VPN, NTP, and SSH. Include relevant screenshots or command outputs.
- Security Features: Describe the security measures you implemented to protect the network.
- **Testing:** Demonstrate how you tested the VPN connection to ensure it is functioning securely. Provide evidence such as successful ping tests or SSH logins.

4. Cybersecurity Threat Assessment (20%)

- **Risk Analysis:** Identify and analyse the main cybersecurity threats to the online gaming industry, such as DDoS attacks or account hacking.
- **Incident Response Plan:** Propose a simple incident response plan that GameSecure Inc. could use to respond to a cyberattack. This should include steps for containment, eradication, and recovery.
- **Recommended Security Technologies:** Recommend and briefly explain any additional security tools or technologies that could help protect the gaming company from cyber threats.

5. Conclusion (10%):

- Summarize your network setup and the cybersecurity threat assessment.
- Discuss how your solutions will help GameSecure Inc. maintain secure operations and protect its customers.